Real-Time Twitter Data Streaming and Sentiments Analysis using Apache Spark

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Data source: Tweets from (Twitter | X)

SAT5165 – Big Data Analytics
Semester Project Presentation

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1. Introduction

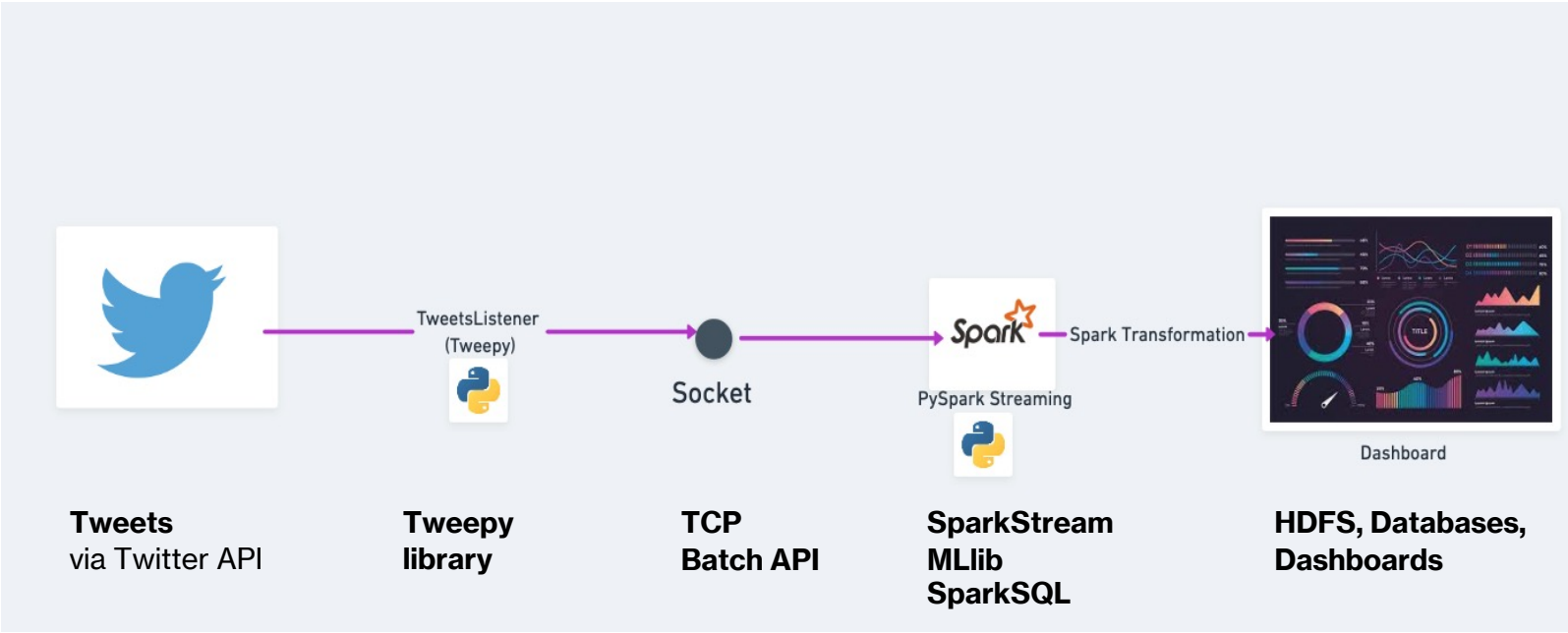
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- **Huge volumes** of data (conversations) exchange on various social media including Twitter (X), Facebook, Tiktok etc. on daily basis.
- **Tracking the content** of such conversations on the fly **can be a daunting task** as they get frequently updated. This initiates a **big data problem**.
- **Apache Spark** can be used track most popular **hashtags** for respective subjects of discussion on Twitter (or other social media platforms) – for real-time data processing and analysis.
- **Applications:** Customer response, sentiments, traffic monitoring etc.

2. Data Streaming Architecture using Spark

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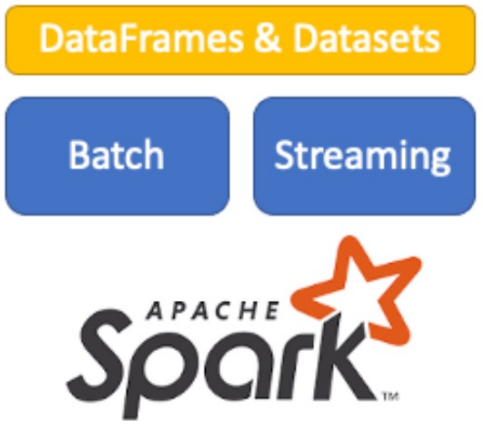




Two Streaming Types

1. Discretizing streaming (D-stream, legacy)

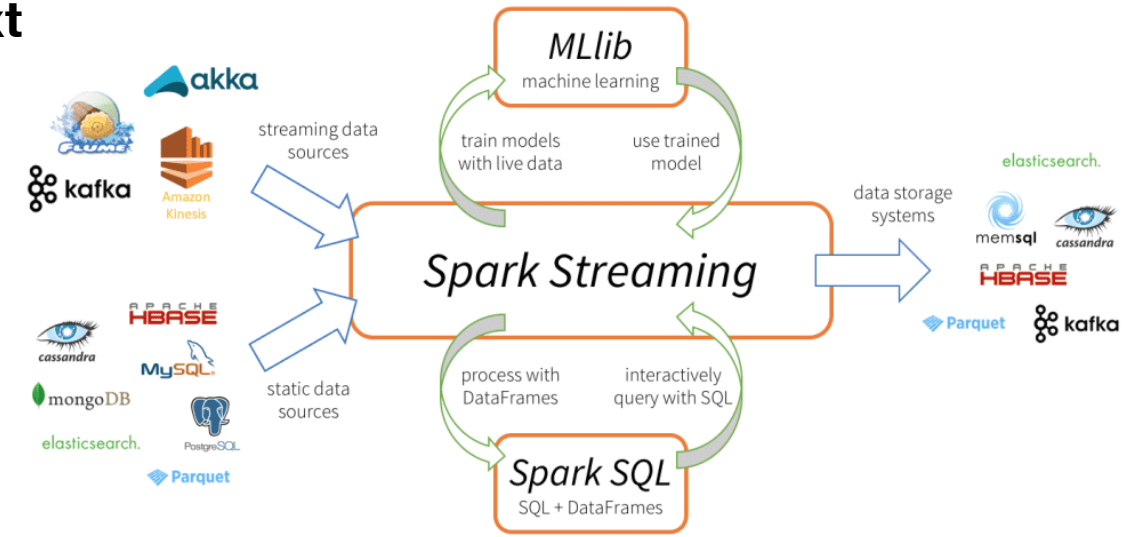
2. Structured streaming



3. Methodology

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- Spark's **StreamingContext** and **SQLContext** can be used to stream data and perform scalable queries in real time.
- To get tweets from Twitter (X): **Access to Twitter API** – consumer key/secret, access tokens and bearer token.

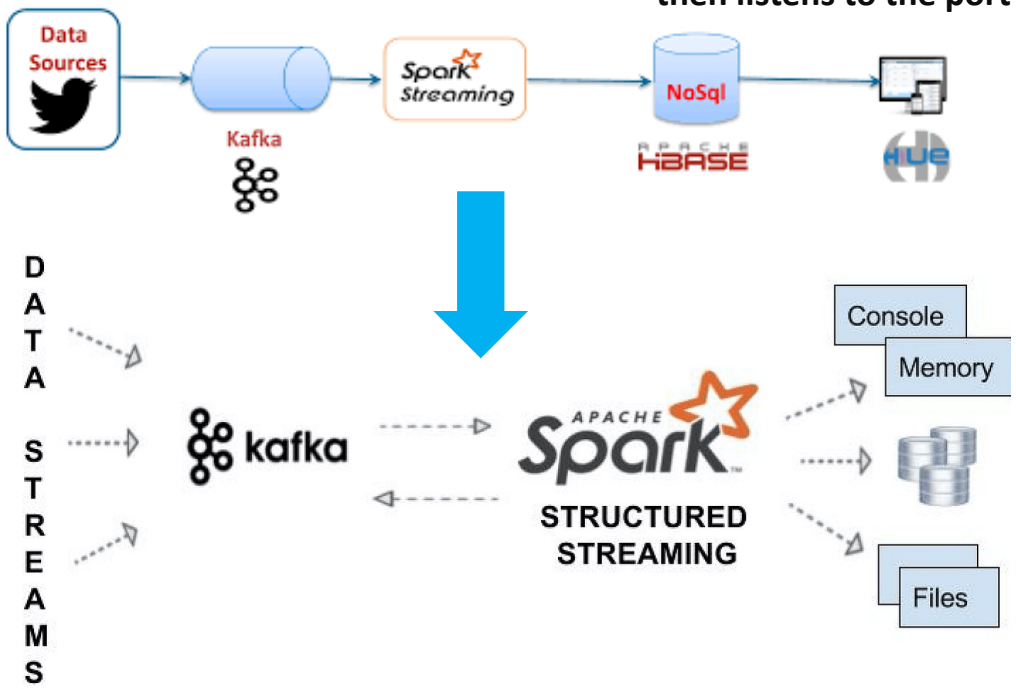


- Python **"Reader.py"** – using the **tweepy** library [nc –lk: push]. Listens to the stream open to the TCP port assigned for streaming data from Twitter.
- Python **"Spark-Session.ipynb"** to implement the Spark model – line by line – **Map**, **Group by Key** and then **Reduce** – returns tweeted words [hash-tagged] and their counts

4. Experiments

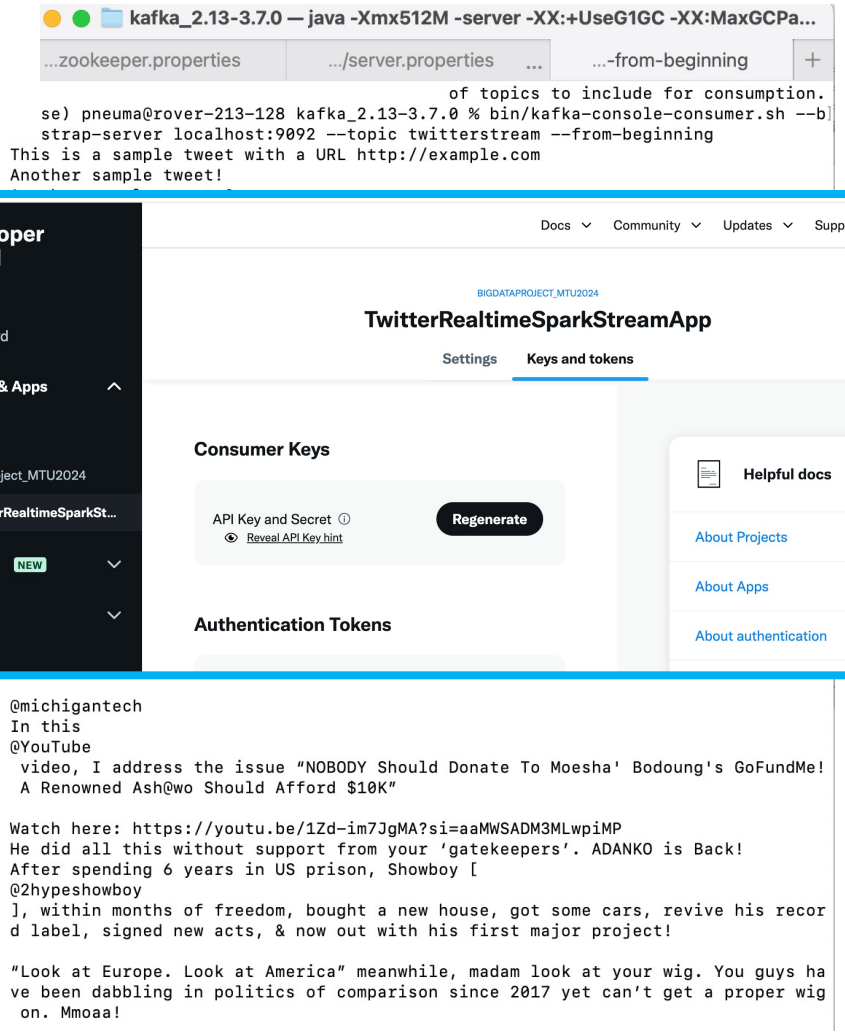
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Twitter (X) no more offers free access to tweets for developers. Used **Netcat to push tweets to local port via TCP/IP protocol. Kafka then listens to the port and queues for Spark**



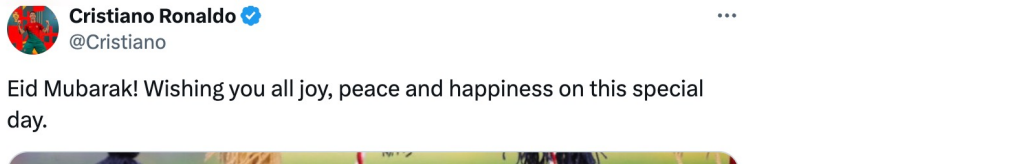
Kafka buffers tweets before processing—via a **distributed queuing service** which can be used to store the data when the **data creation rate is more than processing rate**.

History of streamed data



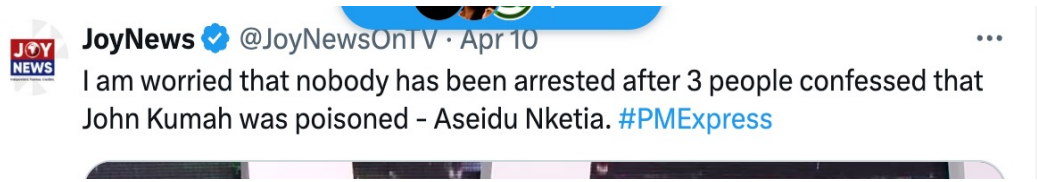
5. Results: Sentiments analysis (+ve, -ve)

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```
SparkSentimentsAnal — python ◂ java -cp ~/bigdatapro/spark-3.5.1-bin-
Batch: 1
-------------------------------------------
24/04/12 02:06:49 INFO CodeGenerator: Code generated in 2.854583 ms
24/04/12 02:06:49 INFO CodeGenerator: Code generated in 3.183666 ms
+---------+-----+
|sentiment|count|
+---------+-----+
| positive|    3|
+---------+-----+
```

Positive




```
24/04/12 02:25:26 INFO WriteToDataSourceV2Exec: Data source write support MicroBatchWrite[epoch:
 5, writer: ConsoleWriter[numRows=20, truncate=true]] is committing.
-------------------------------------------
Batch: 5
-------------------------------------------
+---------+-----+
|sentiment|count|
+---------+-----+
| positive|    3|
| negative|    2|
+---------+-----+
```

Positive



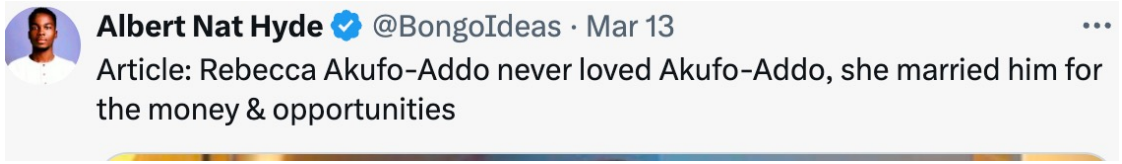

```
24/04/12 02:35:31 INFO WriteToDataSourceV2Exec: Data source write sup
 8, writer: ConsoleWriter[numRows=20, truncate=true]] is committing.
-------------------------------------------
Batch: 8
-------------------------------------------
+---------+-----+
|sentiment|count|
+---------+-----+
| positive|    5|
| negative|    3|
+---------+-----+
```

Positive




```
24/04/12 02:38:25 INFO WriteToDataSourceV2Exec: Data source write supp
 10, writer: ConsoleWriter[numRows=20, truncate=true]] is committing.
-------------------------------------------
Batch: 10
-------------------------------------------
+---------+-----+
|sentiment|count|
+---------+-----+
| positive|    5|
| negative|    5|
+---------+-----+
```

Neutral




```
24/04/12 02:50:27 INFO WriteToDataSourceV2Exec: Data source wri
 18, writer: ConsoleWriter[numRows=20, truncate=true]] is commi
-------------------------------------------
Batch: 18
-------------------------------------------
+---------+-----+
|sentiment|count|
+---------+-----+
| positive|   10|
| negative|   11|
+---------+-----+
```

Negative

6. Results | Conclusions

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- **Apache Spark streaming** makes it easy to build streaming applications and pipelines. This can be used in many areas such as: **monitor sentiments on social media platforms in real-time** and **assess customer response to products and services.**

- **ML models can be implemented in real time.** May not favor cross validation but compensated by the huge volume of data.

References:

Apache Spark – https://spark.apache.org/streaming/
Snowflake – https://quickstarts.snowflake.com/guide/getting_started_with_azure_data_factory_and_snowflake/index.html?index=..%2F..index#7
Apache Kafka – https://kafka.apache.org/
OpenAI (ChatGPT) – https://chat.openai.com/
GitHub Projects -- https://docs.github.com/en/issues/planning-and-tracking-with-projects/learning-about-projects/about-projects
X (Twitter) Developer Projects -- https://developer.twitter.com/en/portal/dashboard

Further work:

- Combine streams from different sources into a repository, then perform **sentiment analysis** on the stream data – in real time.

- **Visualize real-time traffic** using dashboards.

Thank you!